

Sohu
Assignment 4
Spring 25
CSE 350

Instruction:

- ✓ The solution must be in a hand-written format
- ✓ PDF file of the solution needs to be submitted through Google form

Question 1 of 2

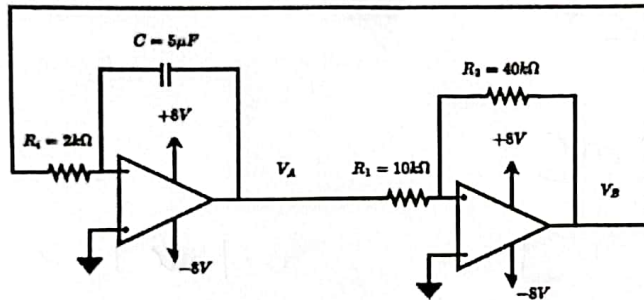


Figure 1: Triangular Wave Generator

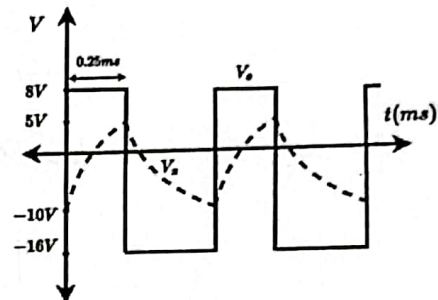


Figure 2: Square Wave Generator Output

(a)	Calculate the time period and the frequency of the triangular wave generator of Figure 1.
(b)	Design a square wave generator that can generate a similar output waveform given in Figure 2. Draw the designed circuit with all necessary values. (You can assume the value of the capacitor 50 nF.)
(c)	Find the duty cycle and frequency of the output waveform of your designed square wave generator.

Question 2 of 2

Truth Table

Input A	Input B	Output
0	0	1
0	1	0
1	0	0
1	1	1

(a)	Design a static CMOS logic circuit that will implement the above truth table.
(b)	For the case A = 0, B = 1, find the state of the all MOS used in your design(On /OFF).

$$1. a) \quad P = \frac{40}{10} = 4$$

symmetrical power supply,

$$\begin{aligned} T &= \frac{4RC}{P} \\ &= \frac{4 \times 2\mu \times 5\mu}{4} \\ &= 10 \mu s \end{aligned}$$

$$f = \frac{1}{T} = \frac{1}{10 \mu s} = 0.1 \text{ kHz}$$

1. b) From plot,

$$V_H = 8V$$

$$V_{TH} = 5V$$

$$V_L = -16V$$

$$V_{TL} = -10V$$

$$T_1 = 0.25 \text{ ms}$$

$$C_x = 50 \text{ nF}$$

$$V_{TH} = V_H \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow 5 = 8 \frac{R_1}{R_1 + R_2}$$

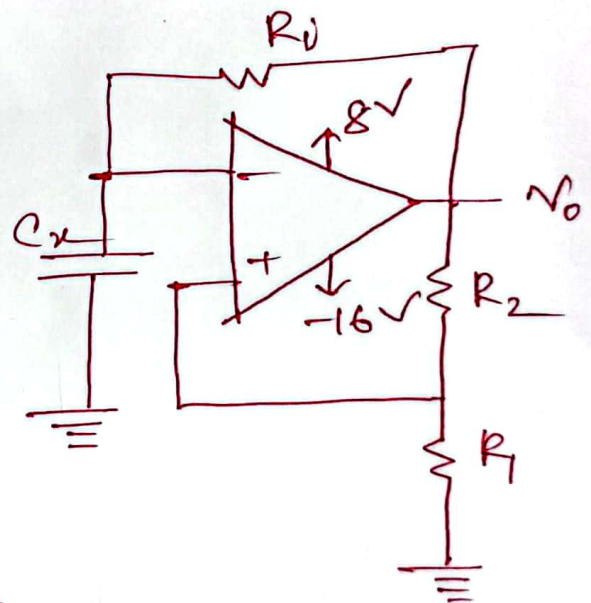
$$5R_1 + 5R_2 = 8R_1$$

$$\Rightarrow 5R_2 = 3R_1$$

$$\text{Let, } R_1 = 10 \text{ k}\Omega,$$

$$5R_2 = 3 \times 10 \text{ k}$$

$$R_2 = 6 \text{ k}\Omega$$



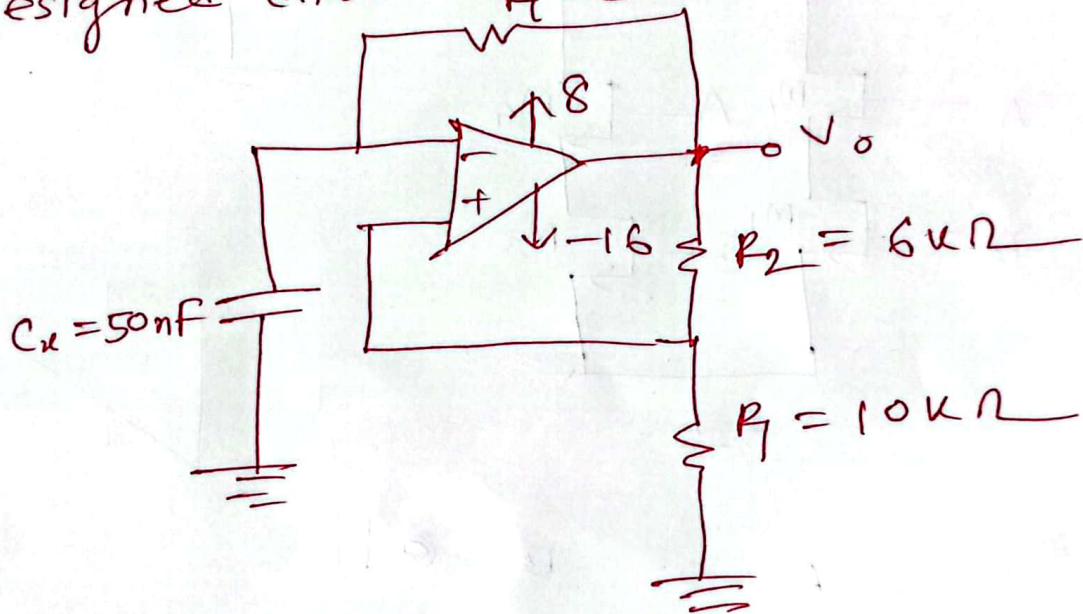
$$T_1 = R_x C_x \ln \frac{V_H - V_L}{V_H - V_{TH}}$$

$$0.25 \times 10^{-3} = R_x \times 50 \times 10^{-9} \ln \frac{8 + 10}{8 - 5}$$

$$\Rightarrow R_x = 2790.55 \Omega$$

$$= 2.79 \text{ k}\Omega$$

Designed circuit: $R_1 = 2.79 \text{ k}\Omega$



1.c) $V_{TH} = 8 \text{ V}$, $V_L = -16 \text{ V}$

$$V_{TH} = 8 \times \frac{10}{10 + 6} = 5 \text{ V}$$

$$V_{TL} = -16 \times \frac{10}{10 + 6} = -10 \text{ V}$$

$$T_1 = 2.79 \times 10^3 \times 50 \times 10^{-9} \ln \frac{8 + 10}{8 - 5}$$

$$= 250 \mu\text{s}$$

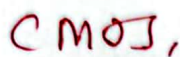
$$T_2 = 2.79 \times 10^3 \times 50 \times 10^{-9} \ln \frac{-16 - 5}{-16 + 10}$$

$$= 175 \mu\text{s}$$

$$f = \frac{1}{T_1 + T_2} = 2.35 \text{ kHz}$$

$$D.C = \frac{T_1}{T_1 + T_2} \times 100\% = \frac{250 \mu\text{s}}{250 \mu\text{s} + 175 \mu\text{s}} \times 100\% = 58.8\%$$

2. a)



2 (b)

$$M_1 \rightarrow \text{off}$$
$$M_2 \rightarrow 0n$$
$$M_3 \rightarrow 0n$$

$M_4 \rightarrow \text{off}$

$$M_5 \rightarrow \infty$$

$m_c \rightarrow 0$

$$M_2 \rightarrow \sigma f J^+$$
$$M_8 \rightarrow \sigma_7$$
$$Y_1 \rightarrow 1$$

$m_g \rightarrow 0$

$$m_{10} \rightarrow \text{off}$$